Nandita Ghosh

B.Tech CSE Student · AI/ML Enthusiast · IoT & Robotics Developer

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Barasat, North 24 Parganas, West Bengal – 700126

Education

B.Tech in Computer Science & Engineering **2024 – Ongoing**

Adamas University, Barasat

Sem 1: 8.45 SGPA **Sem 2:** 9.15 SGPA **Sem 3:** 8.87 SGPA **Sem 4:** 9.45 SGPA **Approx. CGPA: 8.98**

Experience

June–October 2026

Summer Internship [Certificate]

EssProm Academy

- Implemented a project management tool for time and resource tracking , streamlining task allocation and progress monitoring for the team.

Member, Adamas Robotics and AI Club **Feb 2025 – Present**

Adamas University

- Collaborate on IoT, robotics, and AI projects with peers, contributing to hands-on hardware and software builds within the club.

Projects

- **VACUUM-X: Smart Autonomous Vacuum Cleaner**
  Led team project on an autonomous cleaning solution for Indian homes, hospitals, and offices, with real-time dust detection, obstacle-avoidance navigation, mobile app and LED control interface; engineered at Rs. 4K–5K with scalability potential.

- **Smart Helmet: Advanced Rider Safety System**
  Built a dual-sensor accident detection system using accelerometer and gyroscope data, with automatic GPS location transmission to emergency contacts and an algorithm to distinguish real accidents from false triggers, designed to work reliably in low-connectivity areas.

- **ConceptFlow: Number Converting Platform** [ConceptFlow]
  Built a web-based platform, ConceptFlow, that interactively converts numbers between Binary, Decimal, Octal, and Hexadecimal with step-by-step visual animations.

- **IQ Tester: Interactive Intelligence Assessment Platform** [IQ-Tester]
  Developed a web-based IQ assessment platform featuring logical reasoning, pattern recognition, numerical ability, and problem-solving questions with automated scoring and performance evaluation.

Technical Skills

Languages: C++, Python, C, Java, HTML, CSS, JavaScript, React.js, Tailwind CSS

Hardware: Arduino, Raspberry Pi, Microcontroller Programming, Sensor Integration, GPIO Control

Core CS: Data Structures & Algorithms, OOP, Database Design, Full-Stack Development

AI / ML: Machine Learning Fundamentals, Deep Learning Concepts, Data Preprocessing

Tools: Git, GitHub, VS Code, Arduino IDE, Canva

Certification

Basics of Python – Infosys Springboard [Python Course]

Comprehensive training on Python fundamentals: variables, data types, control structures, functions, libraries.

Web Development & AI Workshop – Kshitij, IIT Kharagpur [KTJCertificate]

Participated in a technical workshop covering Web Development and Artificial Intelligence.

Languages

English (Professional) Hindi (Fluent) Bengali (Native)